Akhilesh Rawat

Full-Stack Developer | **Technical Lead** | **Fintech, Lending and ERP Platforms**

Dehradun, India | +91 9119759418 | akhileshrawat.asr@gmail.com
linkedin.com/in/akhilesh-rawat-asr | github.com/Akku-Rawat

Professional Summary

Full-Stack Developer and Technical Lead with 5+ years of experience designing, building and shipping fintech, lending, HR and ERP platforms with React.js, TypeScript, Node.js, Python and Java. Technical Lead for the open-source Tazama case management system and an HRMS built on Frappe, and contributor to the open-source Mojaloop payments platform. Built a Loan Origination System with no-code credit decision engines, Loan Management System modules, a bank transaction reconciliation platform and an ERP integration that passed Zambia Revenue Authority (ZRA) certification. Experienced in system design, microservices, REST and GraphQL APIs, PostgreSQL, Docker, CI/CD and code review.

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, SQL, HTML, CSS
Frontend: React.js, Next.js, Vue.js, Angular, TanStack Query, Zustand, Tailwind CSS, Mantine, Material UI, Bootstrap
Backend: Node.js, Express.js, NestJS, Hono, FastAPI, Django, Flask, Frappe Framework, Spring Boot, Spring, GraphQL, Prisma, Celery, Flowable (BPMN)
Databases: PostgreSQL, MySQL, MongoDB, CouchDB, Oracle Database, Redis, SQLite, Firebase
APIs and Security: RESTful APIs, GraphQL, SOAP, OpenAPI/Swagger, Keycloak, JWT, RBAC, OTP Authentication, AES Encryption
DevOps and Cloud: Docker, Docker Compose, Kubernetes, AWS, GitHub Actions, CI/CD, Linux, Oracle WebLogic, Git, GitHub, GitLab, Postman
Practices: System Design, Microservices, Technical Leadership, Code Review, Unit and Integration Testing, API Design, Database Design
Domain: Loan Origination (LOS), Loan Management (LMS), Frappe HRMS, Payroll and Statutory Compliance, Payments, Fraud and AML, Reconciliation, ERPNext, E-Invoicing (ZRA VSDC), Data Lakehouse, Oracle FLEXCUBE

Work Experience

Software Developer July 2023 – Present
Rolaface Softwares Pvt. Ltd. *Dehradun, India*

Loan Origination System (LOS) | *Python, Frappe, React.js, TypeScript, REST APIs* 2026 – Present

- Designed the UI and built a Loan Origination System end to end on Python/Frappe and React.js, digitizing the full loan journey from application, pre-screening, appraisal and underwriting to e-signed offer and loan booking.
- Built configurable, no-code rule engines for product assignment, pre-screening and eligibility over 20+ applicant variables (income, credit score, DTI, employment, collateral). They automate credit decisions in real time and let business teams change lending policy without code changes.
- Delivered 60+ REST APIs with OpenAPI/Swagger documentation and versioned rule management (effective dates, row-level locking, optimistic concurrency), and set the API standards the team follows.
- Built a configurable approval workflow with role-based assignment, stage transitions, return and rejection, and comments. Every application moves through pre-screening, appraisal, underwriting and offer with a clear record of each decision.
- Built underwriting and offer modules for collateral valuation, legal verification and repayment schedules, with canvas-based e-signature and signed-document upload to close loans digitally.

Loan Management System (LMS) | *Python, Frappe Lending, React.js, TypeScript* 2026 – Present

- Built the core LMS backend on Python/Frappe Lending, with REST APIs for loan products, loan booking, repayment schedules and loan transfers.
- Implemented back-office risk and reporting: arrears tracking, loan classification and provisioning, and loan statements with branded PDF export for audits and customers.
- Built front-office views for loan officers: a portfolio dashboard summarizing loan volumes and applications, and a single loan view showing overview, repayment schedule and instalment status.

HRMS (Technical Lead) | *Python, Frappe HRMS, ERPNext, REST APIs* 2026

- Led the HRMS as Technical Lead and single-handedly built the core backend on Frappe HRMS. It automates onboarding, appraisal and exit workflows, with email notifications for salary slips, upcoming holidays and other HR events.
- Built a configurable company setup (fiscal year, leave policies, holiday lists, weekly offs, shift types) so each organization runs HR on its own rules without code changes.
- Built statutory payroll compliance with configurable salary structures, statutory deductions and user-defined income tax brackets for each country's laws. Payroll runs are automated, with year-to-date totals, bank entries and custom-designed salary slips.

ERP System | *React.js*, *TypeScript*, *Python*, *Frappe*, *ERPNext* 2025 – Present

- **Sales:** Built the sales module end to end: quotations, proforma invoices, sales invoices, credit and debit notes, multi-currency pricing with automatic exchange rates and branded PDF documents.
- **Purchase and inventory:** Built purchase and inventory management covering purchase orders and invoices, imported goods, stock tracking across warehouses, and stock corrections and transfers.
- **Accounting:** Built the accounting module with payment entries and advance allocation, journal entries, receivable and payable aging, and customer and supplier statements.
- **Customers:** Built the customer module with customer groups, credit limits and outstanding balances, plus a single view of each customer's quotations, invoices and aging.
- **Analytics:** Built sales and purchase analytics dashboards that show revenue, spending and outstanding balances.
- **ZRA:** Integrated the ERP with Zambia's statutory tax compliance system (ZRA Smart Invoice) for e-invoicing and stock reporting, passing ZRA certification.

AI-Recon: Bank Transaction Reconciliation Platform | *Python*, *FastAPI*, *Celery*, *Redis*, *PostgreSQL* 2025 – 2026

- Built FastAPI services and a Celery and Redis scheduler that automatically ingest transaction files from 4 source types (HTTP API, FTP, SFTP, database) on cron schedules, with run history and cooldown-based postponement.
- Integrated Oracle FLEXCUBE general ledger and account entries into reconciliation, and built reconciliation run audit APIs with grouped and manual matching across ATM, card, POS, mobile money and wallet channels.
- Built React.js and Mantine screens for reconciliation runs, manual matching, audit logs with PDF reports and scheduler setup, managed schema changes with Alembic, and set up Docker and GitHub Actions CI/CD.

Tazama (Open Source): Case & Investigation Management (Tech Lead) | *NestJS*, *PostgreSQL*, *CouchDB* 2025 – 2026

- Led development as Technical Lead of the case management system for Tazama, an open-source real-time fraud and AML transaction monitoring platform, while writing hands-on backend code with NestJS, Prisma and PostgreSQL. Secured it with Keycloak authentication (JWT) and role-based access control (RBAC), and built filtering, search and sorting for analyst and investigator dashboards.
- Implemented alert triage and the full case lifecycle (assignment, reassignment, suspension, reopening, manual and automated closure) behind REST APIs for alert ingestion and case operations, with strong validation, error handling and consistent state transitions.
- Built case management on the Flowable BPMN workflow engine, routing fraud and AML cases through role-based work queues for investigators, supervisors and compliance teams. Case creation, closure and reopening need a second person's approval (maker-checker), and the flow includes SAR/STR regulatory filing.
- Built the investigation module to securely store KYC, EDD and adverse-media evidence in CouchDB. Every file is encrypted with AES-256-GCM, protected by SHA-256 integrity hashing and file-type and size validation, and access is isolated per tenant.
- Built analytics APIs on the Tazama data lakehouse, writing custom SQL queries for each visualization to power live fraud-monitoring charts such as Benford's Law analysis.

Mojaloop Project (Open Source) | *Node.js*, *GraphQL*, *Prisma*, *MongoDB*, *React.js* 2024 – 2025

- Engineered Prisma-MongoDB schemas and GraphQL APIs for the financial hub reporting service, enabling flexible query capabilities and supporting processing of 100k+ transactions daily.
- Designed React.js reporting hub dashboards integrated with GraphQL APIs, incorporating RBAC and error handling, delivering actionable insights that reduced transaction review time by 30% and highlighting operational pain points for faster decision-making.
- Built DFSP onboarding components with progress tracking, stage-wise status, error indicators, restart functionality, and theme-based UI customization for branding, improving operational efficiency and user experience.
- Achieved 90% test coverage across backend (Node.js) and frontend (React.js) and documented end-to-end workflows, including DFSP onboarding, API integration, and Payment Manager deployment, accelerating team

onboarding.

Backend Developer June 2021 – July 2023
Kaveosoft Innovations Pvt. Ltd. *Haldwani, India*

- Built and secured Spring Boot APIs and microservices with robust testing, API key-secret authentication, and Swagger UI, reducing post-deployment bugs by 40% and accelerating integrations.
- Refactored and optimized legacy APIs and microservices, improving performance 40% and cutting maintenance effort 45%, ensuring stable and efficient backend operations.
- Deployed and managed APIs and financial module services on WebLogic, providing production support, rapid incident resolution, and 99.9% uptime for critical banking systems.

Access South Africa | *Java, Spring Boot, Oracle FLEXCUBE, SOAP, WebLogic* 2023

- Architected and developed enterprise-grade Spring Boot middleware APIs and deployed on WebLogic, transforming high-volume JSON requests into Oracle FLEXCUBE SOAP calls and returning JSON responses, boosting performance by 35%.
- Implemented API key-secret authentication and integrated Swagger UI, enhancing security, accelerating client onboarding, and enabling seamless multi-system integration.

Access Ghana | *Oracle FLEXCUBE, JavaScript, SQL* 2023

- Customized 5+ Oracle FLEXCUBE screens with JavaScript modals to deliver real-time birthday alerts, enabling tellers to provide personalized in-branch customer interactions and improving overall customer engagement.
- Developed optimized SQL queries and robust logic to handle high-volume individual and corporate accounts, accurately managing same-day, belated, and upcoming birthdays across multiple/shared accounts.

Projects

Creddex: Self-Hosted Team Credential Vault | *TypeScript, Node.js, Hono, React.js, SQLite, Docker* 2026

- Built a self-hosted vault for a team's credentials, runbooks and files. Every secret is encrypted at rest with AES-256-GCM, and passwords are hashed with scrypt and checked with timing-safe comparison.
- Implemented role-based access control (viewer, editor, admin), HttpOnly session cookies, rate limiting and a full audit trail of every change, secret reveal, download and sign-in.
- Built a React 19 and TanStack Query UI with Markdown runbooks, drag-and-drop file previews and a Cmd+K command palette, shipped with Docker Compose and a self-resetting demo mode.

Signboard: Website Builder SaaS for Local Businesses | *Next.js, React.js, TypeScript, SQLite, Tailwind CSS* 2026

- Built a SaaS that gives local businesses a mobile-first page (menu or services with prices, hours, photos, location) with WhatsApp ordering and booking, set up through a 4-step onboarding wizard with live preview.
- Implemented phone OTP login (hashed codes, expiry, resend throttling, attempt limits), Free and Pro plan gating, custom domains and 9 data-driven themes rendered with CSS container queries.
- Added SEO (server-side rendering, schema.org structured data, sitemap) and analytics for views, taps and traffic sources from tagged share links and QR codes, excluding bots and owner visits.

Education

Graphic Era Deemed To Be University 2019 – 2023
Bachelor of Technology in Computer Science and Engineering (CGPA: 8.44) *Dehradun, India*